A Mixed-Reality Experimental Indoor Testbed for Evaluation of Drone Flight Control Methods

Benicio B. Cruz * **Jorge A. Ricardo Jr** * **Davi A. Santos** *

* *Mechatronics Department, Aeronautics Institute of Technology (ITA), São José dos Campos, Brazil.*
(corresponding author e-mail: eng.beniciobarbosa@gmail.com)

Abstract:
This paper presents an experimental mixed-reality indoor testbed for the evaluation of drone guidance and control methods. This experimental setup comprises Crazyflie drones, an OptiTrack motion capture system for providing state feedback, and a user-friendly object-oriented MATLAB program structure for the implementation of guidance and control algorithms in real drones or real-time simulated ones. This MATLAB program, besides implementing the physics of the simulated drones and the environment, also allows the user to insert virtual flying obstacles into the scenario. The setup also features a three-dimensional photorealistic visualizer developed in Unity 3D that shows the simulated or real drones along avatars of the virtual obstacles on the same screen. The proposed testbed is illustrated with a collision-free guidance method in two scenarios: one with real drones among virtual obstacles, and another with simulated drones also among virtual obstacles, showing its effectiveness and practicality.

Keywords: autonomous drones; guidance and control; indoor flight experiments; mixed reality.

1. INTRODUCTION

Autonomous systems, particularly in the realm of aerial robotics, have been gaining attention in the last decade. To support academic exploration of guidance and control algorithms for such systems, it is important to develop adaptable and accessible experimental frameworks (Okaya et al., 2023). Within this context, the Crazyflie drone (Bitcraze, 2013) emerges as a pivotal instrument due to its open-source nature and compactness.

Recently, Luis et al. (2020); Soria et al. (2022) have evaluated guidance and control methods using the Crazyflie platform. Luis et al. (2020) proposed a trajectory-tracking control for multiple robots using a model predictive control (MPC) algorithm. The method was initially numerically evaluated in MATLAB based on the Crazyflie dynamics. Subsequently, the method was implemented in C++ and experimentally evaluated using Crazyflie dones. Similarly, Soria et al. (2022) proposed an MPC framework for aerial swarms. The method was numerically evaluated in MATLAB and implemented in Python to be evaluated using Crazyflies. On the other hand, Giernacki et al. (2017) proposed a numerical simulator in MATLAB integrated with a 3D visualizer and implementing the Crazyflie dynamics. It is worth mentioning, that none of the above references have proposed a unique evaluation setup supporting both numerical and experimental evaluations using the Crazyflie drones.

To address the mentioned gap, this paper proposes a mixed-reality experimental indoor testbed for evaluating guidance and control methods. This testbed is based on the integration of a motion capture system for providing pose feedback, a Python script for implementing the communication between the Crazyflies and a desktop computer that implements the outer-loop control and guidance algorithms in an object-oriented MATLAB code, and a photorealistic three-dimensional (3D) visualizer developed in Unity 3D. The setup provides the flexibility to operate with real Crazyflie drones or simulated ones. Moreover, it enables the user to incorporate virtual (simulated) obstacles into the scenario, displaying them among the real or simulated drones in the 3D visualizer, thus justifying the mixed-reality nomenclature. As a result, the setup offers two modes of operation: a pure simulation mode or a mixed-reality simulation mode. In summary, the main contribution of this paper is the proposal of a dual-mode mixed-reality experimental indoor testbed for the evaluation of drone guidance and control methods.

The remaining text is organized in the following manner. Section 2 describes the experimental setup. Section 3 presents results for both numerical and experimental evaluations of a collision-free guidance using the proposed setup. Lastly, Section 4 concludes the paper.

2. SETUP DESCRIPTION

In this section, we present the proposed experimental indoor testbed to support the development and evaluation of drone guidance and control methods. Our approach integrates Crazyflie open-source quadcopters, an OptiTrack motion capture system for pose feedback, and a user-

⋆ Project developed at the Air Robotics Laboratory (LRA) at the Aeronautics Technological Institute (ITA). For more information and access to source code, visit our GitHub repository.
Source code available at: `https://github.com/LRA-ITA/A-Mixed-Reality-Experimental-Indoor-Testbed-for-Evaluation-of-Drone-Flight-Control-Methods.git`

friendly software integration developed using MATLAB, Python, and Unity 3D.

The remaining of this section is organized as follows. Subsection 2.1 introduces the Crazyflie flying platform. Subsection 2.2 briefly describes the OptiTrack camera system. Lastly, Subsection 2.3 presents the proposed experimental setup.

2.1 Crazyflie Drone

The Crazyflie, depicted in Figure 1, is a compact and open-source quadcopter platform, specifically designed for research and education purposes (Bitcraze, 2013). Its lightweight and modular structure, allow for greater adaptability, modification, and experimentation by users and researchers (Giernacki et al., 2017). Additionally, the drone is equipped with onboard sensors, such as an accelerometer and a gyroscope, and allows the integration with external systems capable of providing position information, such as a motion capture positioning system. Moreover, the Crazyflie is equipped with suitable controllers to provide stability and enable the effective tracking of command inputs.

Communication between the Crazyflie and a ground station is established through a radio dongle called Crazyradio (BitCraze, 2013). This communication is necessary for remote control of the drone and data transmission during flight.

2.2 OptiTrack Motion Capture System

The proposed experimental setup uses the OptiTrack motion capture system (OptiTrack, 1996), which is an advanced tracking system equipped with a network of infrared cameras capable of detecting reflective markers, as the one highlighted in Figure 1, positioned on objects. The accuracy of OptiTrack is remarkable due to its high capture frequency, which can reach hundreds of frames per second. In the context of aerial vehicles, this system can accurately observe the position and attitude of the drones, providing this feedback at a suitable frequency for its use in control and navigation algorithms (Soria et al., 2022).

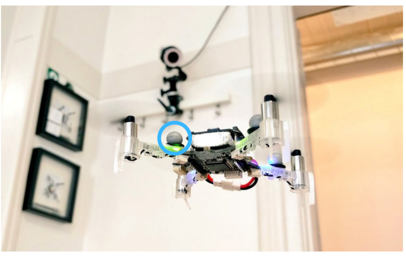

Figure 1. Crazyflie drone with reflective markers.

The data captured by the cameras undergo processing through the Motive optical motion capture software (Zega et al., 2022). This software allows the streaming of the processed data in the local network, enabling its utilization by multiple computers as well as on the same computer running Motive.

2.3 Experimental Setup

This subsection details the proposed dual-mode mixed-reality experimental testbed for drone guidance and control methods. This setup comprises Crazyflie quadcopters, six Optitrack motion capture cameras, and virtual (simulated) obstacles. It uses a user-friendly software integration developed using MATLAB, Python, and Unity 3D, allowing the setup to run in experiment or simulation mode.

In experiment mode, a MATLAB script, designed with a user-friendly object-oriented structure and running in soft real-time, is responsible for guiding the Crazyflies and simulating the obstacles. Moreover, it receives the Crazyflies' states from the cameras through data streaming, communicating both the Crazyflies' states and the guidance commands via a UDP connection to a Python script implementing the Crazyswarm (Preiss et al., 2017) library. This Python script properly processes and transmits this data to the Crazyflies utilizing the Crazyflie radio dongle. Additionally, the MATLAB soft real-time script transmits the poses of the guided Crazyflies and the simulated obstacles to a 3D visualizer coded in Unity 3D. This communication also occurs through a UDP connection.

In simulation mode, the MATLAB script does not integrate with the Crazyflies using the Python script nor the motion capture system to get the Crazyflies' states. Instead, in addition of implementing the guidance of the drones and simulating the obstacles, the MATLAB script also implements the control of the drones and their equations of motion. In this mode, similarly to the experiment mode, it also transmits the poses of the simulated drones and the virtual obstacles to the 3D visualizer.

Figure 2 presents the block diagram of the proposed mixed-reality experimental setup, containing logical switches used to illustrate the two modes of operation. Moreover, Figure 3 shows the 3D visualizer interface, where the black drones are the guided ones and the blue drones represent the obstacles.

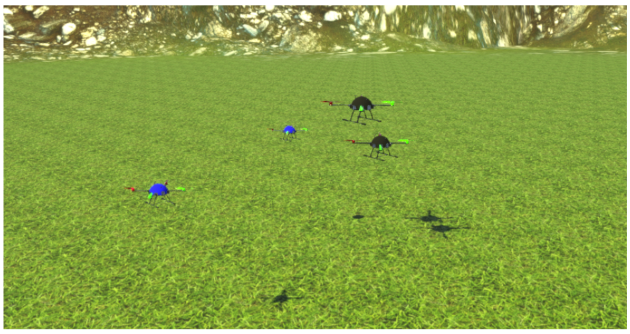

Figure 3. 3D visualizer interface, where the black drones are the guided drones and the blue drones represent the virtual obstacles.

It is worth mentioning that, since the MATLAB script is responsible for the integration between the relevant components for each operation mode, the user can simply change the operation mode by using a flag in the MATLAB script. Moreover, the proposed software integration provides the flexibility to implement the software compo-

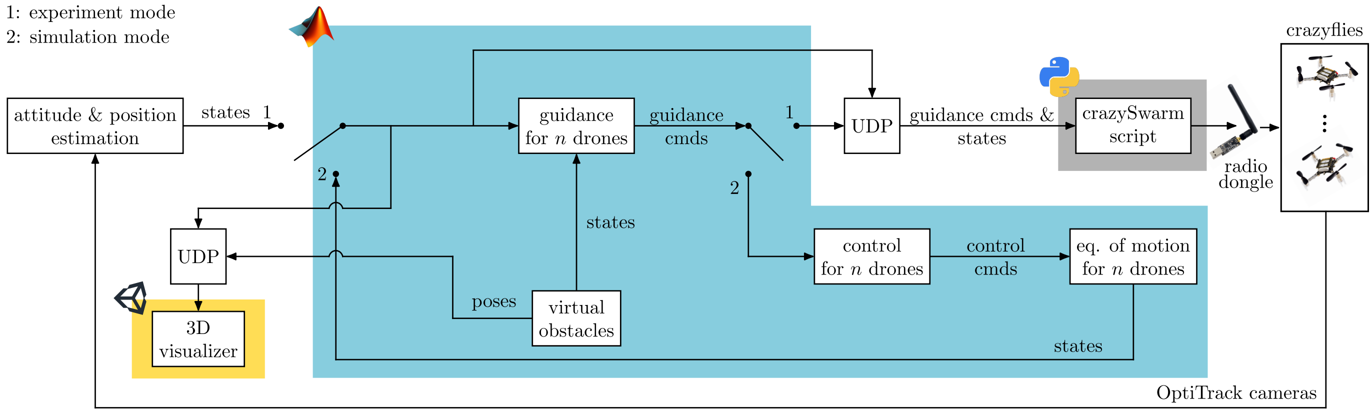

Figure 2. Block diagram of the proposed mixed-reality experimental setup.

nents in different computers operating in the same local network. Figure 4 depicts a schematic of the mixed-reality experimental setup, illustrating the implementation of the software components on the same computer.

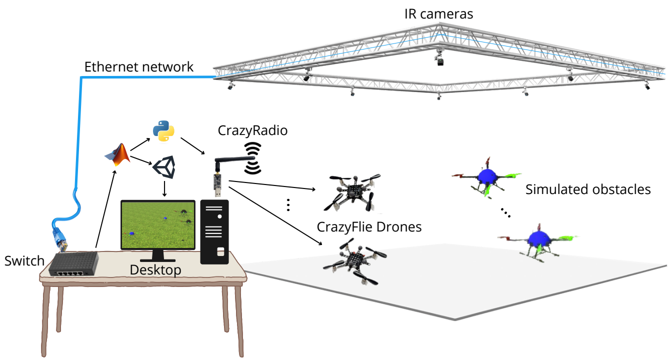

Figure 4. Illustration of the mixed-reality experimental setup.

Another important feature of the proposed setup is that its dual-mode operation allows to evaluate and refine guidance and control techniques before their evaluation in the real experiment. Moreover, it is noteworthy that the possibility to use virtual obstacles along real drones in the experiment mode provides an extra layer of security, as collisions with the virtual obstacles would not damage the real drones.

3. DEMONSTRATION

This section demonstrates the effectiveness and practicality of the proposed setup by implementing a collision-free guidance method (Ricardo and Santos, 2023; Ricardo Jr and Santos, 2023) in both simulation and experiment modes containing virtual obstacles. In this context, Subsection 3.1 and 3.2 presents, respectively, the results obtained in the simulation and experiment modes.

3.1 Simulation Mode

To demonstrate the proposed setup in simulation mode, we have used a scenario containing four guided drones operating in the presence of four virtual obstacles. The guided drones had to follow a sequence of waypoints, from takeoff to landing, while avoiding collisions among them and with the virtual obstacles. Figure 5 depicts the top view of the 3D visualizer for this simulation, captured immediately after the guided drones took off.

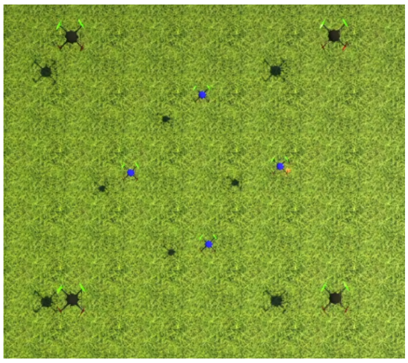

Figure 5. Top view of the 3D visualizer in the proposed simulation. The black drones are the guided ones and the blue drones represent the virtual obstacles.

Figure 6 shows the 3D paths of the drones during the simulation test in four different time instants. To provide a better interpretation and visualization, a video containing the 3D animation of this simulation is available here https://youtu.be/DS8oKmqzaBg.

3.2 Experiment Mode

To demonstrate the proposed setup in experiment mode, we have used a scenario containing two guided Crazyflie real drones operating in the presence of four virtual obstacles. We have conducted two experiments: one where the guided drones followed a sequence of waypoints, from takeoff to landing, defining parallel paths; and another one where the guided drones followed a sequence of waypoints defining conflicting paths. A video showing these experiments, along with their 3D animations is available here https://youtu.be/DS8oKmqzaBg.

Figure 7 shows the 3D paths of the Crazyflies during the first experiment in four different time instants. One can see that the guided Crazyflies successfully executed their paths from takeoff to landing.

Figure 8 shows the 3D paths of the Crazyflies during the second experiment in four different time instants. One can

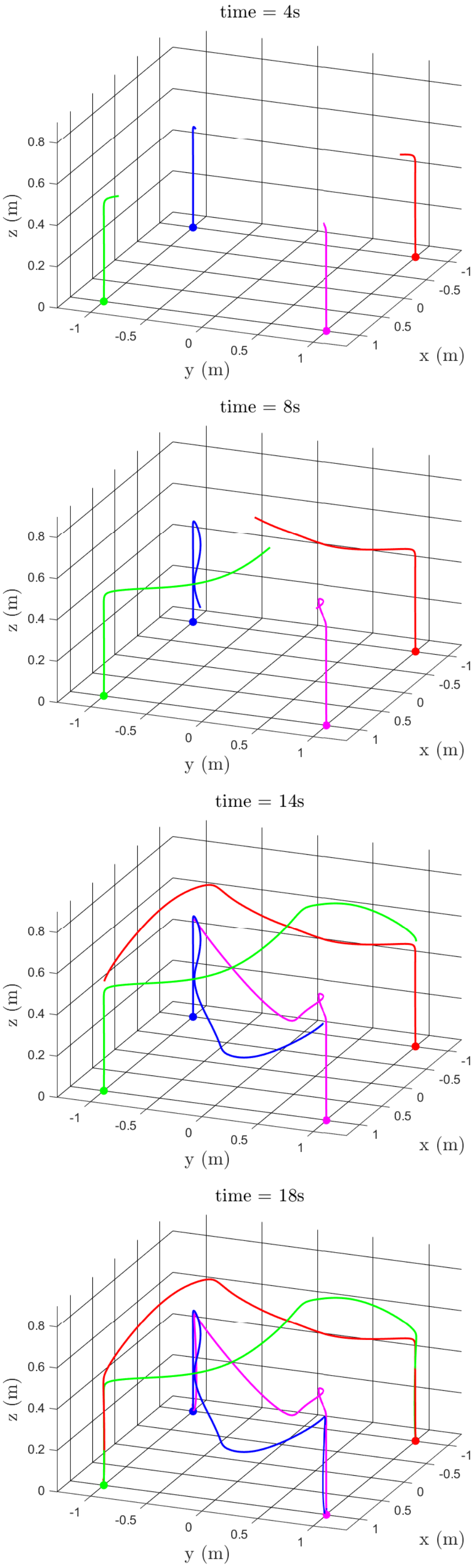

Figure 6. Drone 3D paths in the performed simulation. Legend: — Drone 1 path, — Drone 2 path, — Drone 3 path, — Drone 4 path, and • initial position of the drones.

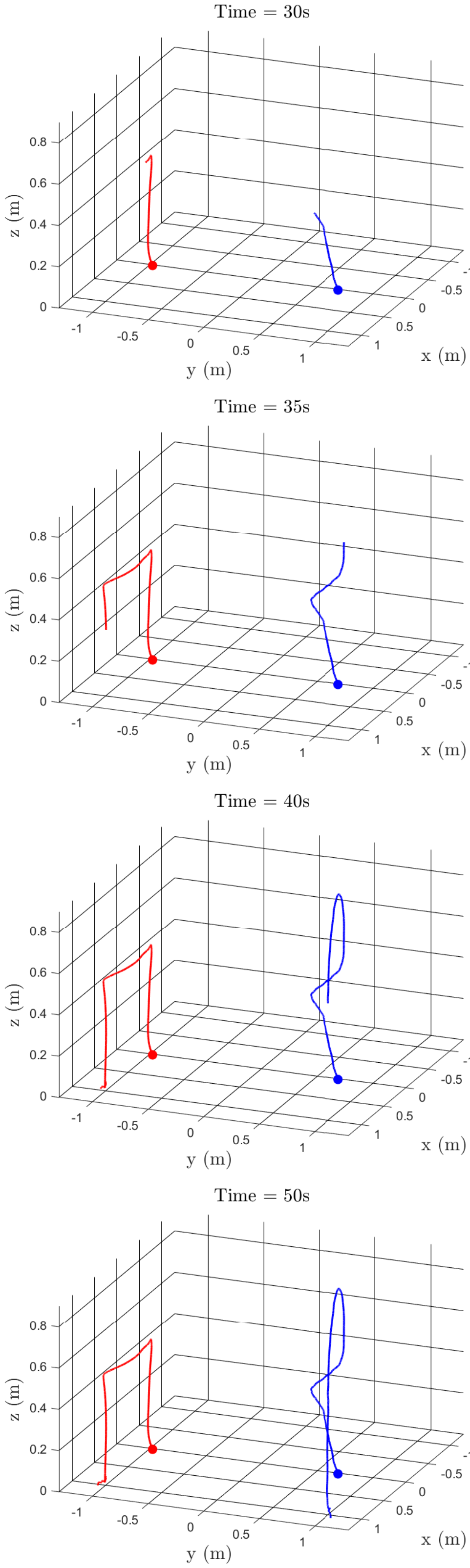

Figure 7. Crazyflies' paths in four different time instants during the first experimental demonstration. Legend: — Drone 1 path, — Drone 2 path, and • initial position of the drones.

see that the guided Crazyflies successfully executed their paths from takeoff to landing while avoiding each other.

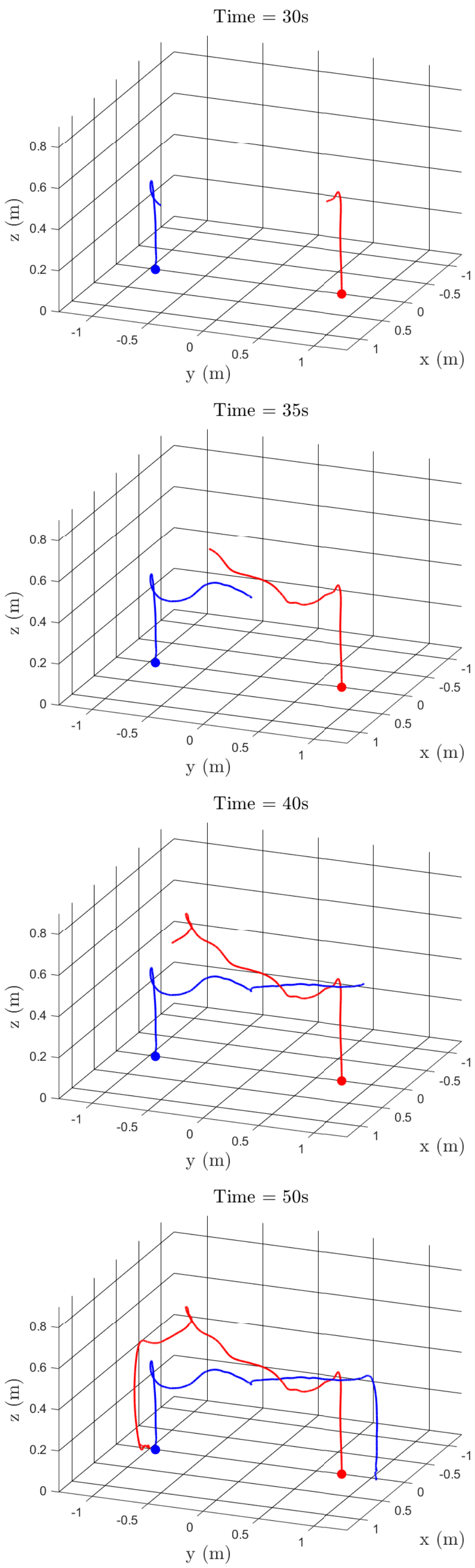

Figure 8. Crazyflies' paths in four different time instants during the second experimental demonstration. Legend: —— Drone 1 path, —— Drone 2 path, and • initial position of the drones.

4. CONCLUSIONS

This paper has proposed a dual-mode mixed-reality experimental indoor testbed for the evaluation of drone flight control methods. The proposed setup comprises Crazyflie drones, an OptiTrack motion capture system, and a user-friendly software integration using MATLAB, Python, and Unity 3D for visualization. The resulting setup allows to evaluate guidance and control methods in both experiment and simulation modes and integrate virtual obstacles into the scenario. The switching between these simulation modes only requires changing a simple flag. Moreover, it uses a user-friendly simple integration between guidance and control methods implemented in MATLAB with Crazyflie drones. The proposed setup has been demonstrated in both simulation and experiment modes implementing a collision-free guidance method, showing to be effective and viable for real-time implementation. The proposed setup has significant potential to support the academic exploration of drone guidance and control algorithms due to its modular framework and simple integration between MATLAB and Crazyflie drones. In future works, the experimental setup can be improved to also support the integration of real drones with simulated ones.

ACKNOWLEDGEMENTS

This work was supported by the Air Robotics Laboratory (LRA), the Aeronautics Institute of Technology (ITA), and the Coordination of Superior Level Staff Improvement (CAPES).

REFERENCES

Bitcraze (2013). Crazyflie 2.0. URL https://www.bitcraze.io/products/old-products/crazyflie-2-0/.

BitCraze (2013). Crazyradio. URL https://www.bitcraze.io/products/crazyradio-2-0/.

Giernacki, W., Skwierczyński, M., Witwicki, W., Wroński, P., and Kozierski, P. (2017). Crazyflie 2.0 quadrotor as a platform for research and education in robotics and control engineering. In *2017 22nd International Conference on Methods and Models in Automation and Robotics (MMAR)*, 37–42. doi:10.1109/MMAR.2017.8046794.

Luis, C.E., Vukosavljev, M., and Schoellig, A.P. (2020). Online trajectory generation with distributed model predictive control for multi-robot motion planning. *IEEE Robotics and Automation Letters*, 5(2), 604–611. doi:10.1109/LRA.2020.2964159.

Okaya, T., Kobayashi, K., and Chujo, W. (2023). A study on indoor positioning of an open-source drone using ar markers. In *2023 VTS Asia Pacific Wireless Communications Symposium (APWCS)*, 1–5. doi:10.1109/APWCS60142.2023.10233961.

OptiTrack (1996). Prime cameras. URL https://optitrack.com/cameras/.

Preiss, J.A., Hönig, W., Sukhatme, G.S., and Ayanian, N. (2017). Crazyswarm: A large nano-quadcopter swarm. In *IEEE International Conference on Robotics and Automation (ICRA)*, 3299–3304. IEEE. doi:10.1109/ICRA.2017.7989376. URL https://doi.org/10.1109/ICRA.2017.7989376. Software available at https://github.com/USC-ACTLab/crazyswarm.

Ricardo, J.A. and Santos, D.A. (2023). Robust collision avoidance for mobile robots in the presence of moving obstacles. *IEEE Control Systems Letters*, 7, 1584–1589. doi:10.1109/LCSYS.2023.3275498.

Ricardo Jr, J.A. and Santos, D.A. (2023). Robust collision-free guidance and control for underactuated multirotor aerial vehicles. *Drones*, 7(10), 611. doi:10.3390/drones7100611.

Soria, E., Schiano, F., and Floreano, D. (2022). Distributed predictive drone swarms in cluttered environments. *IEEE Robotics and Automation Letters*, 7(1), 73–80. doi:10.1109/LRA.2021.3118091.

Zega, S.A., Saropi, A., Ronaldy, T., Ilhami, Y., and Farabi, M. (2022). Mengatasi data error pada proses data cleaning motion capture motive optitrack. *JOURNAL OF APPLIED MULTIMEDIA AND NETWORKING*, 6(1), 89–94.